

OBJECT ORIENTED PROGRAMMING USING REAL TIME APPLICATIONS (DSA0135)

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1. Student Analytics System

```
// STUDENT ANALYTICS SYSTEM
#include <iostream>
using namespace std;

class Student
{
private:
    string name;
    int marks;

public:
    void input() { cin >> name >> marks; }
    int getMarks() { return marks; }
    string getName() { return name; }
};

class Department
{
private:
    string deptName;
    int n;

public:
    Student s[100];

    void input()
    {
        cout<<"Department Name: ";
        cin>>deptName;
        cout<<"Number of Students: ";
        cin>>n;

        for(int i=0;i<n;i++)
            s[i].input();
    }

    void topper()
    {
        if(n==0)
        {
            cout<<"Empty Department\n";
            return;
        }

        int maxMarks=s[0].getMarks();
        string topper=s[0].getName();

        for(int i=1;i<n;i++)
        {
            if(s[i].getMarks()>maxMarks)
            {
                maxMarks=s[i].getMarks();
                topper=s[i].getName();
            }
        }

        cout<<"Topper: "<<topper<<" "<<maxMarks<<endl;
    }

    float passPercentage()
    {
        if(n==0) return 0;

        int pass=0;
```

```

        for(int i=0;i<n;i++)
            if(s[i].getMarks()>=40) pass++;
        return (pass*100.0)/n;
    }
};

int main()
{
    int d;
    cin>>d;

    Department dept[10];

    for(int i=0;i<d;i++) dept[i].input();

    for(int i=0;i<d;i++)
    {
        dept[i].topper();
        cout<<"Pass % = "<<dept[i].passPercentage()<<endl;
    }
    return 0;
}

```

2. Election Result Analysis System

```
// ELECTION RESULT ANALYSIS SYSTEM
#include<iostream>
using namespace std;

class Election
{
public:
    static int totalVotes;
    static int maxVotes;
    static int winnerIndex;

    void process(int votes[][50], int regions, int candidates)
    {
        int total[50] = {0};

        for(int j = 0; j < candidates; j++)
        {
            for(int i = 0; i < regions; i++)
            {
                total[j] += votes[i][j];
                totalVotes += votes[i][j];
            }

            if(total[j] > maxVotes)
            {
                maxVotes = total[j];
                winnerIndex = j;
            }
        }

        cout << "Total Votes = " << totalVotes << endl;
        cout << "Winner Candidate Index = " << winnerIndex + 1 << endl;
        cout << "Max Votes = " << maxVotes << endl;
    }
};

int Election::totalVotes = 0;
int Election::maxVotes = 0;
int Election::winnerIndex = 0;

int main()
{
    int regions, candidates;
    cin >> regions >> candidates;

    int votes[50][50];

    for(int i = 0; i < regions; i++)
        for(int j = 0; j < candidates; j++)
            cin >> votes[i][j];

    Election e;
    e.process(votes, regions, candidates);

    return 0;
}
```

3. Store Sales Management System

```
// STORE SALES MANAGEMENT SYSTEM
#include<iostream>
using namespace std;

class Product
{
public:
    string name;
    float price;
    int qty;

    void input()
    {
        cin >> name >> price >> qty;
    }

    float sales()
    {
        return price * qty;
    }
};

class Store
{
public:
    Product p[100];
    int n;

    static float overallSales;
    static string bestProduct;
    static float maxProductSales;

    void input()
    {
        cin >> n;
        for(int i = 0; i < n; i++)
            p[i].input();
    }

    float calculateStoreSales()
    {
        float total = 0;

        for(int i = 0; i < n; i++)
        {
            float s = p[i].sales();
            total += s;

            if(s > maxProductSales)
            {
                maxProductSales = s;
                bestProduct = p[i].name;
            }
        }

        overallSales += total;
        return total;
    }
};

float Store::overallSales = 0;
float Store::maxProductSales = 0;
string Store::bestProduct = "";

int main()
{
    int stores;
    cin >> stores;

    Store s[50];

    float bestStoreSales = 0;
    int bestStoreIndex = 0;
    for(int i = 0; i < stores; i++)
```

```
{
    s[i].input();

    float total = s[i].calculateStoreSales();

    if(total > bestStoreSales)
    {
        bestStoreSales = total;
        bestStoreIndex = i;
    }
}

cout << "Best Store: " << bestStoreIndex + 1 << endl;
cout << "Best Selling Product: " << Store::bestProduct << endl;
cout << "Overall Sales: " << Store::overallSales << endl;

return 0;
}
```